

Reflection Report – Customer Shopping Behavior Analysis

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Track: 2 – GitHub Data Analysis Project

1. Analytical Problem and Intended User

The analytical problem is: what factors significantly influence customers' single-transaction purchase amount? Understanding purchase drivers is a fundamental business question that applies across retail industries. The intended users are marketing teams in small-to-medium retail enterprises who need data-driven insights for targeted promotions, advertising budget allocation, and customer segmentation. By identifying which demographic and behavioural factors correlate with higher spending, marketing teams can move from generic campaigns to personalised strategies.

2. Dataset and Selection Justification

I used the Customer Shopping Behavior Dataset, accessed on 15 April 2026 from a public repository. It contains 3,900 records with 18 fields covering customer demographics (age, gender), purchase details (product category, amount, season, discount usage), and transaction information (payment method, purchase frequency). I selected this dataset for three reasons. First, its structure mirrors real-world retail transaction logs, making the analysis directly relevant to the stated business problem. Second, it includes both categorical and numerical variables, allowing me to demonstrate a full pandas-based analytical workflow. Third, the data are already clean and well-formatted, which is appropriate for the module level and allowed me to focus on meaningful analysis rather than spending excessive time on data rescue.

3. Methodological Approach

Following the assignment guidance, I used Python with pandas exclusively, adding only matplotlib for visualisation. My workflow proceeded through four stages. First, data loading and initial inspection using `pd.read_csv()`, `df.info()`, and `df.describe()` to understand data structure and quality. Second, data cleaning and preparation: checking for missing values (none found), removing duplicates, and creating an age-group categorical variable (18–30, 31–45, 46–60, 60+) to enable segmented analysis. Third, exploratory analysis using pandas `groupby()` operations to compute average purchase amounts across gender, age group, payment method, purchase frequency, and product category. Fourth, visualisation using matplotlib to create bar charts, box plots, and pivot tables that communicate findings clearly. The analysis is substantive but not overly

complex – this is intentional, as the assignment rubric emphasises that “simple but well-executed analysis is preferable to an over-ambitious product that you do not understand.”

4. Key Findings and Interpretation

The analysis revealed several meaningful patterns. First, female customers spend more than male customers on average across all age groups. Second, the 31–45 age group is the highest-spending segment, followed by the 46–60 group, suggesting that mid-career professionals have greater disposable income. Third, payment method correlates with spending: Venmo and PayPal users have higher average transaction values than cash or credit card users, which may reflect younger, more digitally engaged customers. Fourth, purchase frequency is positively associated with average transaction value – weekly shoppers spend more per trip than annual shoppers, indicating that loyalty programmes could be effective. Fifth, among product categories, electronics consistently leads in average purchase amount, while accessories rank lowest. Finally, cross-analysis shows that females in the 31–45 age group have the highest spending among all gender-age combinations.

5. Limitations and Professional Practice

Several limitations should be acknowledged. The dataset is synthetic, so patterns may not generalise to real retail environments. The analysis uses only descriptive statistics; no inferential tests were conducted to confirm whether observed differences are statistically significant. The sample size (3,900 records) is relatively small, and no temporal information is available to analyse seasonal or promotional effects. In professional practice, I would complement this analysis with hypothesis testing (e.g., t-tests or ANOVA) and a larger, real-world dataset before making operational recommendations. I would also incorporate regression analysis to quantify the relative impact of each factor on purchase amount.

6. AI Disclosure

I used AI tools as learning and development aids. The following describes my specific usage.

- DeepSeek (DeepSeek-V3, accessed 15 April 2026): Provided the initial Jupyter Notebook code structure, including data loading, cleaning, groupby aggregations, and visualisations. Helped debug several errors encountered during execution: `NameError` (missing `import matplotlib`), and `FileNotFoundError` (missing figures folder). Supplied step-by-step guidance on using Jupyter Notebook on Mac, writing `README.md`.

- Claude 3.5 Sonnet (accessed 14 April 2026): Explained pandas groupby and agg functions with examples. Suggested the age-group categorisation logic using a custom function. Recommended matplotlib visualisation best practices. Helped me understand how to resolve the FileNotFoundError by creating the figures directory.

My own contribution: I chose the analytical problem and the dataset. I understood every line of code before running it. I manually inspected the actual column names using `df.columns` and corrected all references. I ran the notebook multiple times, tested different grouping combinations, and decided which findings to highlight. I interpreted all outputs and formulated the business recommendations based on my own understanding.